

## -- 1) VIEWS

```
CREATE OR REPLACE VIEW vw_product AS
SELECT *
FROM product;
```

```
CREATE OR REPLACE VIEW vw_product_maior_preco AS
SELECT *
FROM product
WHERE unitprice = (SELECT MAX(unitprice) FROM product);
```

```
CREATE OR REPLACE VIEW vw_product_menor_preco AS
SELECT *
FROM product
WHERE unitprice = (SELECT MIN(unitprice) FROM product);
```

```
CREATE OR REPLACE VIEW vw_media_unitprice AS
SELECT AVG(unitprice) AS media_unitprice
FROM product;
```

```
CREATE OR REPLACE VIEW vw_product_insert AS
SELECT id, productname, supplierid, unitprice, package, isdiscontinued
FROM product;
```

```
INSERT INTO vw_product_insert
(id, productname, supplierid, unitprice, package, isdiscontinued)
VALUES
(78, 'Produto Teste', 1, 25.00, '10 unidades', 0);
```

```
CREATE OR REPLACE VIEW vw_customer_telefone_55 AS
SELECT *
FROM customer
WHERE phone LIKE '55%';
```

## -- 2) STORED PROCEDURES

```
DELIMITER //
```

```
CREATE PROCEDURE listar_produtos()
BEGIN
    SELECT *
    FROM product
    ORDER BY productname;
END //
```

```
CREATE PROCEDURE produtos_menor_20()
BEGIN
```

```
SELECT *
FROM product
WHERE unitprice < 20.00;
END //
```

```
CREATE PROCEDURE executar_todas()
BEGIN
    CALL listar_produtos();
    CALL produtos_menor_20();
END //
```

```
DELIMITER ;
```

```
-- 3) TABELA DE LOG
```

```
CREATE TABLE log_dml (
    id INT AUTO_INCREMENT PRIMARY KEY,
    data_hora DATETIME,
    tabela VARCHAR(50),
    operacao VARCHAR(20),
    usuario VARCHAR(100),
    valor_antigo TEXT,
    valor_novo TEXT
);
```

```
-- 4) TRIGGERS DA TABELA PRODUCT
```

```
DELIMITER //
```

```
CREATE TRIGGER trg_product_insert
AFTER INSERT ON product
FOR EACH ROW
BEGIN
    INSERT INTO log_dml
    (data_hora, tabela, operacao, usuario, valor_antigo, valor_novo)
    VALUES
    (
        NOW(),
        'product',
        'INSERT',
        CURRENT_USER(),
        NULL,
        CONCAT(
            'id=', NEW.id,
            ', productname=', NEW.productname,
            ', supplierid=', NEW.supplierid,
```

```

        ', unitprice=', NEW.unitprice,
        ', package=', NEW.package,
        ', isdiscontinued=', NEW.isdiscontinued
    )
);
END //

```

```

CREATE TRIGGER trg_product_update
AFTER UPDATE ON product
FOR EACH ROW
BEGIN
    INSERT INTO log_dml
    (data_hora, tabela, operacao, usuario, valor_antigo, valor_novo)
    VALUES
    (
        NOW(),
        'product',
        'UPDATE',
        CURRENT_USER(),
        CONCAT(
            'id=', OLD.id,
            ', productname=', OLD.productname,
            ', supplierid=', OLD.supplierid,
            ', unitprice=', OLD.unitprice,
            ', package=', OLD.package,
            ', isdiscontinued=', OLD.isdiscontinued
        ),
        CONCAT(
            'id=', NEW.id,
            ', productname=', NEW.productname,
            ', supplierid=', NEW.supplierid,
            ', unitprice=', NEW.unitprice,
            ', package=', NEW.package,
            ', isdiscontinued=', NEW.isdiscontinued
        )
    );
END //

```

```

CREATE TRIGGER trg_product_delete
AFTER DELETE ON product
FOR EACH ROW
BEGIN
    INSERT INTO log_dml
    (data_hora, tabela, operacao, usuario, valor_antigo, valor_novo)
    VALUES
    (
        NOW(),
        'product',

```

```

        'DELETE',
        CURRENT_USER(),
        CONCAT(
            'id=', OLD.id,
            ', productname=', OLD.productname,
            ', supplierid=', OLD.supplierid,
            ', unitprice=', OLD.unitprice,
            ', package=', OLD.package,
            ', isdiscontinued=', OLD.isdiscontinued
        ),
        NULL
    );
END //

```

-- 5) TRIGGERS DA TABELA CUSTOMER

```

CREATE TRIGGER trg_customer_insert
AFTER INSERT ON customer
FOR EACH ROW
BEGIN
    INSERT INTO log_dml
    (data_hora, tabela, operacao, usuario, valor_antigo, valor_novo)
    VALUES
    (
        NOW(),
        'customer',
        'INSERT',
        CURRENT_USER(),
        NULL,
        CONCAT(
            'id=', NEW.id,
            ', firstname=', NEW.firstname,
            ', lastname=', NEW.lastname,
            ', city=', NEW.city,
            ', country=', NEW.country,
            ', phone=', NEW.phone
        )
    );
END //

```

```

CREATE TRIGGER trg_customer_update
AFTER UPDATE ON customer
FOR EACH ROW
BEGIN
    INSERT INTO log_dml
    (data_hora, tabela, operacao, usuario, valor_antigo, valor_novo)
    VALUES

```

```

(
    NOW(),
    'customer',
    'UPDATE',
    CURRENT_USER(),
    CONCAT(
        'id=', OLD.id,
        ', firstname=', OLD.firstname,
        ', lastname=', OLD.lastname,
        ', city=', OLD.city,
        ', country=', OLD.country,
        ', phone=', OLD.phone
    ),
    CONCAT(
        'id=', NEW.id,
        ', firstname=', NEW.firstname,
        ', lastname=', NEW.lastname,
        ', city=', NEW.city,
        ', country=', NEW.country,
        ', phone=', NEW.phone
    )
);
END //

CREATE TRIGGER trg_customer_delete
AFTER DELETE ON customer
FOR EACH ROW
BEGIN
    INSERT INTO log_dml
    (data_hora, tabela, operacao, usuario, valor_antigo, valor_novo)
    VALUES
    (
        NOW(),
        'customer',
        'DELETE',
        CURRENT_USER(),
        CONCAT(
            'id=', OLD.id,
            ', firstname=', OLD.firstname,
            ', lastname=', OLD.lastname,
            ', city=', OLD.city,
            ', country=', OLD.country,
            ', phone=', OLD.phone
        ),
        NULL
    );
END //

```

DELIMITER ;

-- 6) EXECUTAR AS PROCEDURES

CALL listar\_produtos();

CALL produtos\_menor\_20();

CALL executar\_todas();

-- 7) MOSTRAR OS OBJETOS CRIADOS

SHOW FULL TABLES

WHERE Table\_type = 'VIEW';

SHOW PROCEDURE STATUS

WHERE Db = DATABASE();

SHOW TRIGGERS;

SHOW FUNCTION STATUS

WHERE Db = DATABASE();